

terminals. These could be low-end machines running Atom-like processors or even old PCs that were heading for the scrap heap because they could not run some bloated proprietary OS. All that these terminals (now functioning as thin clients) need is the capability to boot from a network. What's more: such a set-up also allows you to interface Wintel machines from that part of your set-up that needs a proprietary computing environment.

The advantages of this computing architecture include: easy software updates (you only require to update the server/s), potentially lower power requirements, and enhanced security. All this thanks to the Linux Terminal Server Project (LTSP).

Check out the success stories at ltsp.org for what real people are doing with such set-ups. Also, try *K12LTSPorg* for a Linux distribution that bundles LTSP with CentOS, and *K12Linux.org* for a Fedora distribution that bundles LTSP with it.

A software application called iTALC enables you to control all terminals from a central master workstation in the case of a large deployment of terminals. It allows, for instance, a teacher to give a demo from her terminal and make it appear on all student terminals.

The FOSS world around Linux also offers other technologies to facilitate virtualisation, enhanced security, spam protection, desktop publishing, and a host of other business functions, all in a virus-free environment with a range of very functional and attractive graphical desktops. This by itself would make FOSS a strong contender to run a school's entire IT infrastructure rather than just the computer lab. But wait till you learn about what else FOSS has to offer for a learning environment.

Information management

Even small- to medium-sized schools can reap the benefits of information management. FOSS offers some sensible options here too. The list is not complete, and I might have missed your favourite application.

SchoolTool (schooltool.org) calls itself a student information system. It can be



Figure 1: Koha catalogue at DPL

deployed at the school level to track attendance, grades, demographics, timetables, and calendars. It does not cover accounting. It can also be used by individual teachers on their laptops to track the classes they are teaching, for instance. SchoolTool is backed by the Shuttleworth Foundation (shuttleworthfoundation.org), which works closely with the South African school system to promote FOSS.

OpenSIS (opensis.com) is a comprehensive SIS that could easily rival commercial offerings. The base system comes free. Additional modules and support can be purchased. There is a hosted service too, in case schools do not want to host their own system. However, the basic system appears quite adequate to give a clear productivity boost to a school's processes.

OnlineGrades (onlinegrades.org) is not really an online grade-book, but helps host information on grades in a secure manner for controlled access by students and teachers.

Openadmin for Schools is another SIS for schools

